



DALHOUSIE UNIVERSITY

CSCI 5408: Data Management, Warehousing and Analytics

Assignment: Lab - 04

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GitLab Assignment Link: <https://git.cs.dal.ca/parva/>

CSCI5408_W25_B01026328_ParvaMineshbhai_Patel

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1) Created Tables: User, Wallet, Transaction

DDL Queries

```
CREATE DATABASE lab4_transactions;
```

```
USE lab4_transactions;
```

```
CREATE TABLE User_Details (  
    user_id INT PRIMARY KEY AUTO_INCREMENT,  
    full_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) NOT NULL UNIQUE,  
    phone VARCHAR(15) NOT NULL UNIQUE,  
    registration_date DATE NOT NULL,  
    verification_status ENUM('verified', 'unverified') NOT NULL  
);
```

```
CREATE TABLE Wallet_Details (  
    wallet_id INT PRIMARY KEY AUTO_INCREMENT,  
    user_id INT NOT NULL,  
    current_balance DECIMAL(10, 2) NOT NULL DEFAULT 0.00,  
    wallet_status ENUM('active', 'inactive') NOT NULL,  
    FOREIGN KEY (user_id) REFERENCES User_Details(user_id)  
);
```

```
CREATE TABLE Transaction_Details (  
    transaction_id INT PRIMARY KEY AUTO_INCREMENT,  
    sender_wallet_id INT NOT NULL,
```

```

receiver_wallet_id INT NOT NULL,

amount DECIMAL(10, 2) NOT NULL,

timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

transaction_type ENUM('p2p', 'merchant') NOT NULL,

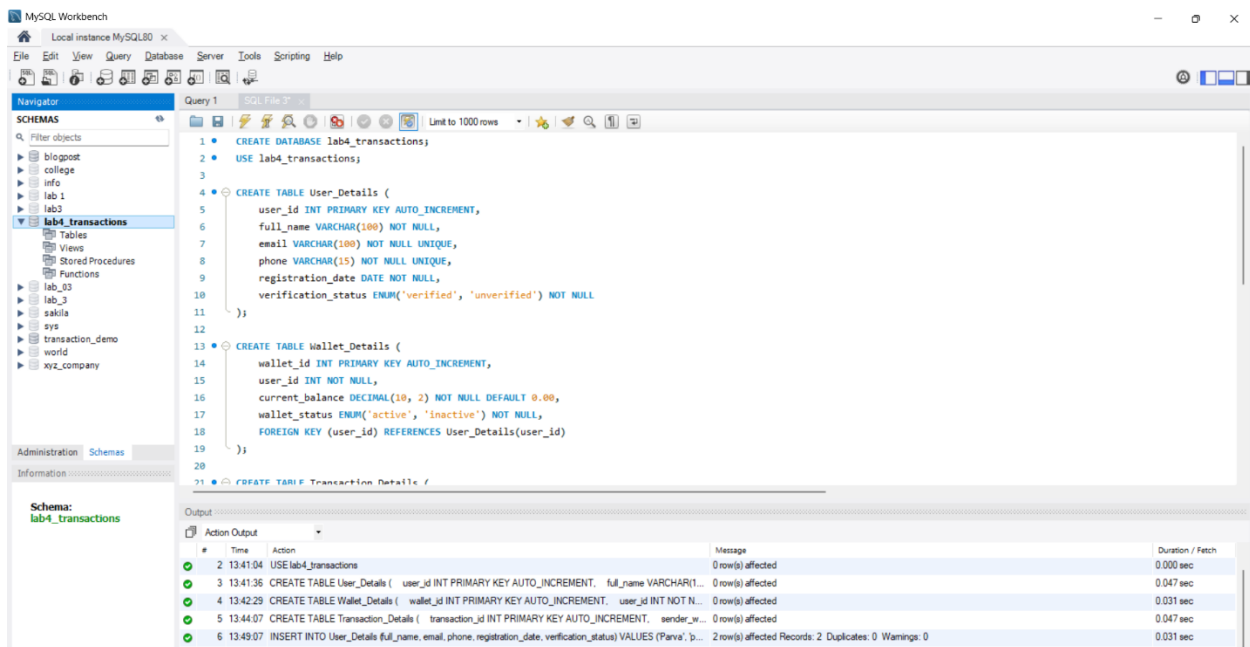
transaction_status ENUM('pending', 'completed', 'failed') NOT NULL,

FOREIGN KEY (sender_wallet_id) REFERENCES Wallet_Details(wallet_id),

FOREIGN KEY (receiver_wallet_id) REFERENCES Wallet_Details(wallet_id)

);

```



Created Tables

2) Inserted Sample Data

DML Queries

```
INSERT INTO User_Details (full_name, email, phone, registration_date, verification_status)

VALUES

('Parva', 'parva2612@gmail.com', '9028182094', '2025-01-01', 'verified'),

('Dev', 'dev2105@gmail.com', '9452930981', '2025-01-02', 'verified');
```

```
INSERT INTO Wallet_Details (user_id, current_balance, wallet_status)

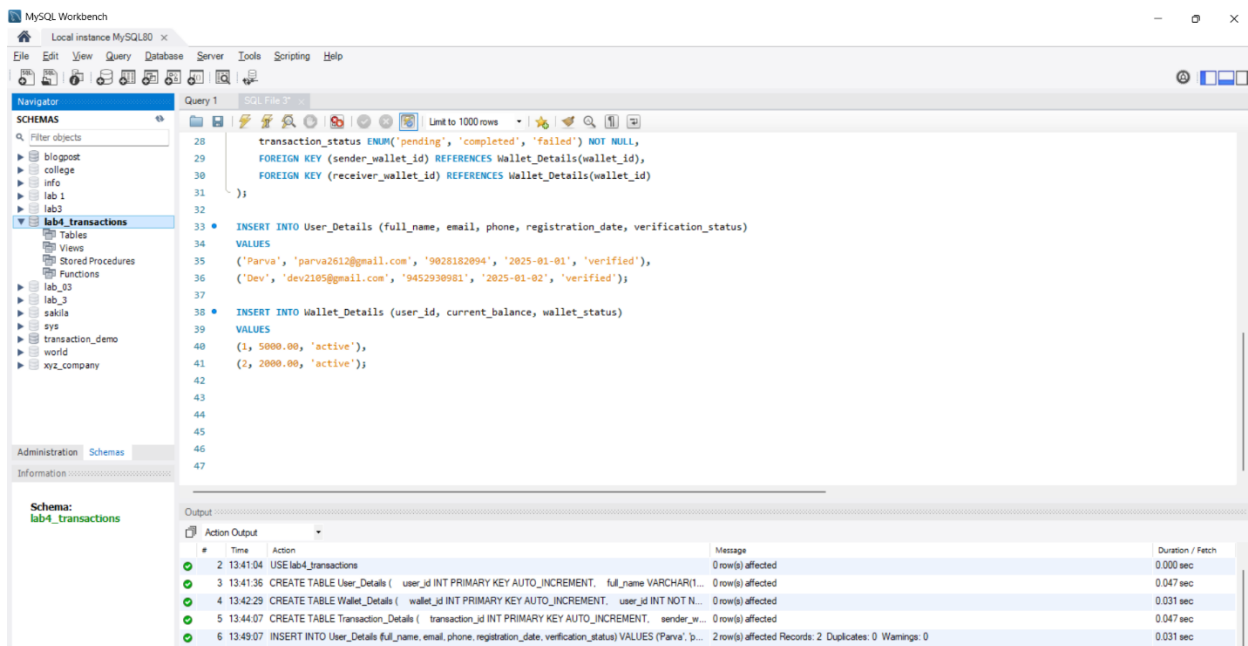
VALUES

(1, 5000.00, 'active'),

(2, 2000.00, 'active');
```

```
select * from User_Details;

select * from Wallet_Details;
```



Inserted Sample Data

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Schema: lab4_transactions

Query 1

SQL File 3

Limit to 1000 rows

```

36 ('Dev', 'dev2105@gmail.com', '9452930981', '2025-01-02', 'verified');
37
38 INSERT INTO Wallet_Details (user_id, current_balance, wallet_status)
39 VALUES
40 (1, 5000.00, 'active'),
41 (2, 2000.00, 'active');
42
43 select * from User_Details;
44

```

Result Grid

	user_id	full_name	email	phone	registration_date	verification_status
1	Parva	parva2612@gmail.com	9028182094	2025-01-01	verified	
2	Dev	dev2105@gmail.com	9452930981	2025-01-02	verified	

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	13:41:36	CREATE TABLE User_Details (user_id INT PRIMARY KEY AUTO_INCREMENT, full_name VARCHAR(100), email VARCHAR(100), phone VARCHAR(20), registration_date DATE, verification_status VARCHAR(20))	0 row(s) affected	0.047 sec
4	13:42:29	CREATE TABLE Wallet_Details (wallet_id INT PRIMARY KEY AUTO_INCREMENT, user_id INT NOT NULL, current_balance DECIMAL(10,2), wallet_status VARCHAR(20))	0 row(s) affected	0.031 sec
5	13:44:07	CREATE TABLE Transaction_Details (transaction_id INT PRIMARY KEY AUTO_INCREMENT, sender_id INT NOT NULL, receiver_id INT NOT NULL, amount DECIMAL(10,2), transaction_date DATE, transaction_status VARCHAR(20))	0 row(s) affected	0.047 sec
6	13:49:07	INSERT INTO User_Details (full_name, email, phone, registration_date, verification_status) VALUES (Parva, 'parva2612@gmail.com', '9028182094', '2025-01-01', 'verified')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.031 sec
7	13:49:36	INSERT INTO Wallet_Details (user_id, current_balance, wallet_status) VALUES (1, 5000.00, 'active'), (2, 2000.00, 'active')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec

Display User_Details Table

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Schema: lab4_transactions

Query 1

SQL File 3

Limit to 1000 rows

```

38 INSERT INTO Wallet_Details (user_id, current_balance, wallet_status)
39 VALUES
40 (1, 5000.00, 'active'),
41 (2, 2000.00, 'active');
42
43 select * from User_Details;
44
45 select * from Wallet_Details;
46

```

Result Grid

	wallet_id	user_id	current_balance	wallet_status
1	1	1	5000.00	active
2	2	2	2000.00	active

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	13:42:29	CREATE TABLE Wallet_Details (wallet_id INT PRIMARY KEY AUTO_INCREMENT, user_id INT NOT NULL, current_balance DECIMAL(10,2), wallet_status VARCHAR(20))	0 row(s) affected	0.031 sec
5	13:44:07	CREATE TABLE Transaction_Details (transaction_id INT PRIMARY KEY AUTO_INCREMENT, sender_id INT NOT NULL, receiver_id INT NOT NULL, amount DECIMAL(10,2), transaction_date DATE, transaction_status VARCHAR(20))	0 row(s) affected	0.047 sec
6	13:49:07	INSERT INTO User_Details (full_name, email, phone, registration_date, verification_status) VALUES (Parva, 'parva2612@gmail.com', '9028182094', '2025-01-01', 'verified')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.031 sec
7	13:49:36	INSERT INTO Wallet_Details (user_id, current_balance, wallet_status) VALUES (1, 5000.00, 'active'), (2, 2000.00, 'active')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
8	13:52:45	select * from User_Details LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

Display Wallet_Details Table

Explanation

The DDL queries create a database called **lab4_transactions** and created three tables such as **User_Details**, **Wallet_Details**, and **Transaction_Details**, with relationships between them using foreign keys. The **User_Details** table stores user information, the **Wallet_Details** table holds wallet information linked to users, and the **Transaction_Details** table tracks transactions between wallets. The DML queries add sample data for users and their wallets, and the **SELECT** statements are used to display the data that has been inserted.

3) Transaction Scenario-1: Transaction "Completed" state

Queries

```
USE lab4_transactions;
```

```
SET AUTOCOMMIT = 0;
```

```
START TRANSACTION;
```

```
UPDATE Wallet_Details
```

```
SET current_balance = current_balance - 1000.00
```

```
WHERE wallet_id = 1;
```

```
SAVEPOINT hold_balance;
```

```
INSERT INTO Transaction_Details (sender_wallet_id, receiver_wallet_id, amount,  
transaction_type, transaction_status)
```

```
VALUES (1, 2, 1000.00, 'p2p', 'pending');
```

```
SAVEPOINT transaction_pending;
```

```
UPDATE Wallet_Details
```

```
SET current_balance = current_balance + 1000.00
```

```
WHERE wallet_id = 2;
```

```
SAVEPOINT credit_wallet;
```

```
UPDATE Transaction_Details
```

```
SET transaction_status = 'completed'
```

```
WHERE transaction_id = LAST_INSERT_ID();
```

COMMIT;

SELECT * FROM Wallet_Details;

SELECT * FROM Transaction_Details;

Explanation

1. **Start Transaction:** The transaction is initiated with **START TRANSACTION**, meaning the changes made will be grouped together and committed at once.
2. **Deduct Amount from Parva's Wallet:** The balance of Parva's wallet (**wallet_id 1**) is reduced by 1000.00 to hold the amount for the transaction.
3. **Savepoint after Holding Parva's Balance:** A **savepoint** named **hold_balance** is created, marking the point after Parva's wallet balance has been deducted, allowing for a potential rollback.
4. **Insert Transaction Record:** A record is inserted into the **Transaction_Details** table with the transaction details, marking the status as pending.
5. **Savepoint after Inserting Transaction Record:** Another **savepoint** named **transaction_pending** is created, allowing for rollback after inserting the transaction record.
6. **Credit Dev's Wallet:** Dev's wallet (**wallet_id 2**) is credited with 1000.00, completing the transaction.
7. **Savepoint after Credit to Dev's Wallet:** A **savepoint** named **credit_wallet** is created after Dev's wallet is credited, enabling rollback if needed.
8. **Update Transaction Status to Completed:** The transaction status is updated to completed, and all the changes are committed, finalizing the transaction.

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Information

Table: wallet_details

Columns:

wallet_id int AI PK
user_id int
current_balance decimal(10,
wallet_status enum('acti

Query 1 Step-1 Create_tables Transaction_Completed

Limit to 1000 rows

```
22 • SAVEPOINT credit_wallet;  
23  
24 • UPDATE Transaction_Details  
25   SET transaction_status = 'completed'  
26   WHERE transaction_id = LAST_INSERT_ID();  
27  
28 • COMMIT;  
29  
30 • SELECT * FROM Wallet_Details;
```

Result Grid

wallet_id	user_id	current_balance	wallet_status
1	1	4000.00	active
2	2	3000.00	active

Output

Action Output

#	Time	Action	Message	Duration / Fetch
16	14:10:08	SAVEPOINT transaction_pending	0 row(s) affected	0.000 sec
17	14:10:26	UPDATE Wallet_Details SET current_balance = current_balance + 1000.00 WHERE wallet_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
18	14:10:44	SAVEPOINT credit_wallet	0 row(s) affected	0.000 sec
19	14:11:05	UPDATE Transaction_Details SET transaction_status = 'completed' WHERE transaction_id = LAST_INSERT_ID()	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
20	14:11:20	COMMIT	0 row(s) affected	0.016 sec

Amount Transferred

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Information

Table: wallet_details

Columns:

wallet_id int AI PK
user_id int
current_balance decimal(10,
wallet_status enum('acti

Query 1 Step-1 Create_tables Transaction_Completed

Limit to 1000 rows

```
24 • UPDATE Transaction_Details  
25   SET transaction_status = 'completed'  
26   WHERE transaction_id = LAST_INSERT_ID();  
27  
28 • COMMIT;  
29  
30 • SELECT * FROM Wallet_Details;  
31  
32 • SELECT * FROM Transaction_Details;
```

Result Grid

transaction_id	sender_wallet_id	receiver_wallet_id	amount	timestamp	transaction_type	transaction_status
1	1	2	1000.00	2025-02-02 14:09:23	p2p	completed

Output

Action Output

#	Time	Action	Message	Duration / Fetch
17	14:10:26	UPDATE Wallet_Details SET current_balance = current_balance + 1000.00 WHERE wallet_id = 2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
18	14:10:44	SAVEPOINT credit_wallet	0 row(s) affected	0.000 sec
19	14:11:05	UPDATE Transaction_Details SET transaction_status = 'completed' WHERE transaction_id = LAST_INSERT_ID()	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
20	14:11:20	COMMIT	0 row(s) affected	0.016 sec
21	14:15:13	SELECT * FROM Wallet_Details LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

Transaction Completed

4) Transaction Scenario-2: Transaction “Failed” state

Queries

```
USE lab4_transactions;
```

```
SET AUTOCOMMIT = 0;
```

```
START TRANSACTION;
```

```
UPDATE Wallet_Details
```

```
SET current_balance = current_balance - 500.00
```

```
WHERE wallet_id = 1;
```

```
SAVEPOINT hold_balance;
```

```
INSERT INTO Transaction_Details (sender_wallet_id, receiver_wallet_id, amount,  
transaction_type, transaction_status)
```

```
VALUES (1, 2, 500.00, 'p2p', 'pending');
```

```
SAVEPOINT transaction_pending;
```

```
ROLLBACK TO hold_balance;
```

```
UPDATE Transaction_Details
```

```
SET transaction_status = 'failed'
```

```
WHERE transaction_id = LAST_INSERT_ID();
```

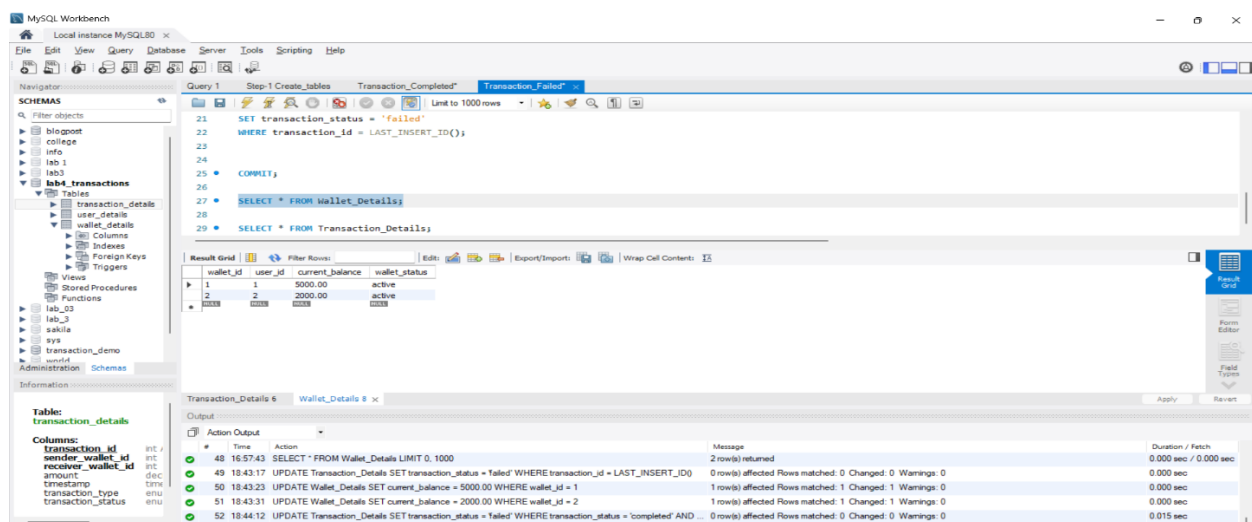
```
COMMIT;
```

```
SELECT * FROM Wallet_Details;
```

```
SELECT * FROM Transaction_Details;
```

Explanation

1. **Start Transaction:** The transaction begins with **START TRANSACTION**, grouping the operations for atomic execution.
2. **Deduct Amount from Parva's Wallet:** Parva's wallet (**wallet_id 1**) balance is reduced by 500.00, temporarily holding the amount for the transaction.
3. **Savepoint after Holding Parva's Balance:** A **savepoint** named **hold_balance** is created, allowing rollback to this point if needed.
4. **Insert Transaction Record:** A record is inserted into the **Transaction_Details** table, indicating a pending transaction of 500.00 from Parva to Dev.
5. **Savepoint after Inserting Transaction Record:** A savepoint named **transaction_pending** is created, marking the point after the transaction record insertion.
6. **Rollback to Hold Balance Savepoint:** The transaction rolls back to the **hold_balance** savepoint, reversing the deduction from Parva's wallet and canceling the transaction.
7. **Update Transaction Status to Failed:** The transaction status is updated to failed, reflecting that the transaction was not completed.
8. **Commit Changes:** The rollback and status update are committed, finalizing the transaction as failed.



No Transaction in table

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- transaction_demo
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- Administration
- Schemas

Query 1 Step-1 Create tables Transaction_Completed Transaction_Failed

Limit to 1000 rows

```
17
18 ROLLBACK TO hold_balance;
19
20 UPDATE Transaction_Details
21 SET transaction_status = 'failed'
22 WHERE transaction_id = LAST_INSERT_ID();
23
24 COMMIT;
25
```

Result Grid

transaction_id	sender_wallet_id	receiver_wallet_id	amount	timestamp	transaction_type	transaction_status
1	1	2	1000.00	2025-02-02 14:09:23	p2p	completed
3	1	2	500.00	2025-02-02 14:34:57	p2p	failed

Table: transaction_details

Columns:

- transaction_id int /
- sender_wallet_id int /
- receiver_wallet_id int /
- amount dec
- timestamp tmu
- transaction_type enu
- transaction_status enu

Transaction_Details 5 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
39	14:34:52	SAVEPOINT hold_balance	0 row(s) affected	0.000 sec
40	14:34:57	INSERT INTO Transaction_Details (sender_wallet_id, receiver_wallet_id, amount, transaction_type, transac...	1 row(s) affected	0.000 sec
41	14:35:04	SELECT * FROM Transaction_Details LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
42	14:36:39	SELECT * FROM Wallet_Details LIMIT 0, 1000	2 row(s) returned	0.015 sec / 0.000 sec
43	14:36:57	UPDATE Transaction_Details SET transaction_status = 'failed' WHERE transaction_id = LAST_INSERT_ID()	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

Transaction Failed

5) References

1. GeeksforGeeks, "SQL Transactions - COMMIT, ROLLBACK, and SAVEPOINT," GeeksforGeeks [Online], Available: <https://www.geeksforgeeks.org/sql-transactions/> [Accessed: February 02, 2025].
2. Oracle Corporation, "MySQL Workbench - Database Design & Modeling Tool," MySQL [Software], Available: <https://www.mysql.com/products/workbench/>. [Accessed: February 02, 2025].